

df10.1

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1 Modules

Let R be a ring, $S < R$ be a subring. Then R is a S -module. Let R be a ring and $M < R$ be an additive subgroup. Then M is an R -module iff M is a left ideal of R . If R is a ring, then $R[X]$ is an R -module, $R^{n \times n}$ is an R -module.

If M, N are R -modules, then $M \times N$ is an R -module, with the ring action given by $r(m, n) = (rm, rn)$.

If M is an R -module and $N < M$ is a submodule, then M/N is an R -module. The subtlety is from $r(m+N) = rm+N$ being well-defined. If $m+N = m'+N$, then $m-m' \in N$, so $r(m-m') \in N$, so $rm+N = rm'+N$.

Let M be an R -module and define $N < M$ to be the right annihilator of $A \subset R$ by $rn = 0$ for all $r \in A$, so N is indeed a subgroup of M . If R is commutative, then N is a submodule of M .

$(\mathbb{R}^3, +, \times)$, where $\times$ is cross-product, is not a ring.

Let R be a ring, G a finite group, so the group-ring RG is an R -module.

Can any (finite dimension) matrix ring be embedded in a group ring?

Let V/K be a vector space. Then V has a basis. Let $A \subset V$ be called independent if for all finite $\{v_i\} \subset A$ we have $\sum k_i v_i = 0 \rightarrow (k_i) = \vec{0}$. Denote by $\mathcal{A}$ the set of all independent subsets of V and let $\mathcal{C} \subset \mathcal{A}$ be a chain. I claim $A = \bigcup \mathcal{C}$ is independent. Suppose $\{v_i\} \subset A$. Pick $v_i \in A_\alpha$ for some maximal α (since there are finitely many i 's). Then if $\sum k_i v_i = 0$, we have $(k_i) = \vec{0}$. By Zorn's lemma, let $M \in \mathcal{A}$ be a maximal independent set.

$\text{Sps FTSOC Span}(M) \subsetneq V$. Pick $v \in V - \text{Span}(M)$ and consider $\sum k_i v_i + kv = 0$. If $k = 0$, then $(k_i) = 0$. Else, $v = \sum (-k^{-1}k_i)v_i \in \text{Span}(M)$, cont. Then $M \cup \{v\}$ is an independent set, cont max of M .

Let K^J be the vector space st if $x \in K^J$, $x_\alpha \neq 0$ for finitely many $\alpha \in J$.

If $x, y \in K^J$, then $x + y, kx \in K^J$. If $V \cong K^J$ for some J , is J unique up to cardinality?

What is an R -algebra. Why is R commutative with identity?